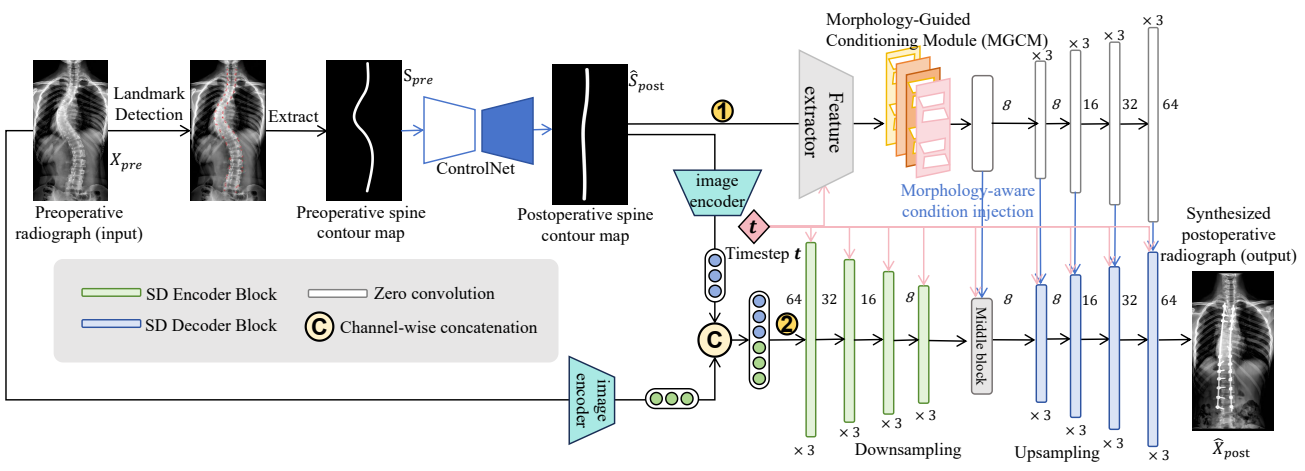

Landmark Detection
X_{pre}
Extract
S_{pre}
ControlNet
$\hat{S}_{post}$
Preoperative radiograph (input)
Preoperative spine contour map
Postoperative spine contour map
Morphology-Guided Conditioning Module (MGCM)
Feature extractor
Morphology-aware condition injection
image encoder
Timestep t
Middle block
Downsampling
Upsampling
Synthesized postoperative radiograph (output)
$\widehat{X}_{post}$
SD Encoder Block
SD Decoder Block
Zero convolution
Channel-wise concatenation